

Prajwal Kumar K

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[LinkedIn](#) | [Portfolio](#) | [GitHub](#) | [Codeforces](#) | [CodeChef](#) | [LeetCode](#) | [AtCoder](#)

EDUCATION

Indian Institute of Information Technology, Kottayam

Kottayam, Kerala

Bachelor of Technology in Computer Science and Engineering; CGPA: 8.66/10

Aug 2024 – May 2028 (expected)

TECHNICAL SKILLS

Programming: C++, Rust, Python, TypeScript, JavaScript, C, Java, SQL, Solidity, MATLAB

AI/ML: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Matplotlib

Research and engineering: Jupyter, Simulink, FastAPI, Next.js

Research interests: Algorithms and systems, game theory, machine learning and NLP, AI safety

Languages: English (Fluent), German (Business Proficiency), Hindi, Kannada

EXPERIENCE

Coders Club Coordinator

Jan 2025 – Present

IIIT Kottayam

Kottayam, Kerala

- Organize programming contests with 80+ participants; curate problem sets and configure Codeforces Gym rounds.

Openverse Technical Lead (Open Source Club)

May 2025 – Jul 2026

IIIT Kottayam

Kottayam, Kerala

- Led technical work and PR reviews for 50+ Openverse contributors; authored Algorithmica sorting visualizations.

Full-Stack Web Developer Intern

Dec 2025 – Mar 2026

CIO Mogul (Magazine)

Remote

- Delivered 3 magazine sites in a 2-person team—[CIO Mogul](#), [CIO Mogul India](#), and [CIO Healthcare](#)—for publishing articles and digital issues.
- Automated PDF cover extraction, page navigation, and A3-spread splitting across 3 publications using Next.js/React and Supabase/PostgreSQL editorial workflows.

PROJECTS

PokeForge | *Python, PyTorch, Gen 9 OU*

[Code](#)

- Prototyped a Gen 9 OU Pokémon agent evaluation testbed with 2 pinned baselines (in progress).
- Integrated pinned FoulPlay/Metamon baselines; 61 offline tests cover selected action, state, and damage contracts while simulator conformance remains under evaluation.

Bluff | *Imperfect-information games, opponent modeling*

[Code](#)

- Designed a 2-player imperfect-information card game to study bluff detection and opponent adaptation.
- Implemented Bayesian opponent profiles and a learned public-information detector; improved draw-aware score by 4.9–5.8 percentage points across 8 synthetic profiles versus detector ablation.

Drishti-XAI | *Retinal imaging, MATLAB, Python*

- Evaluated an explainable retinal-screening prototype for generalization across 2 diabetic-retinopathy image datasets.
- Built the MATLAB/Python prototype and evaluation pipeline; at a fixed model and threshold, sensitivity/specificity were 0.908/0.877 on APTOS (n=733) and 0.363/0.865 on Messidor-2 (AUC 0.666).

SiliconSpire | *C++, Rust, Metaheuristics*

[C++](#) | [Rust](#)

- Engineered a Quadratic Assignment Problem solver to minimize weighted flow-distance cost for facility placement.
- Combined Grey Wolf Optimization and Tabu Search; reached published optima in 77/90 Rust runs (85.6%) versus 34/90 C++ runs across 3 small QAPLIB instances.

BattleCP | *Rust, Axum, Tokio, WebSockets*

[Live](#) | [Code](#)

- Developed a multiplayer Battleship game that awards turns for solving Codeforces problems.
- Implemented Rust/Axum WebSockets, Codeforces validation, server-side rules, and reconnects; recorded 6,922 cumulative page views.

ACHIEVEMENTS AND CERTIFICATIONS

Publication: First-listed author and presenter, “Collaboration-Enabled Energy-Aware Learning Model Exchange (CE-LME) Framework for Edge Computing-Assisted Applications,” IEEE ICICI 2025. [Paper](#).

Algorithmica: selected for and presented at OOSC 3.0, IIT Kanpur (2025); team presentation at T4E 2025, IIT Madras.

Hackathon: Won Best Software Award at IEEE MACE .hack(); 2026 with Team Hogorithm (Anchor).

Competitive programming: Codeforces Pupil (peak 1221), CodeChef peak 1463, 400+ problems solved across platforms.

Cyfrin Updraft (2025): Blockchain Basics, Rust Programming Basics, Solidity Smart Contract Development, and Fundamentals of Zero-Knowledge Proofs.